Blood Group Prediction using Fingerprint Pattern Analysis

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Introduction & Problem Statement

Introduction

- Fingerprints are unique biometric patterns formed before birth.
- They remain unchanged throughout life.
- Ridge and minutiae features are used for identification.
- Fingerprint patterns may relate to blood group.
- Deep learning can analyze these patterns for prediction.

Problem Statement

- Traditional blood group detection requires blood samples and lab testing.
- The process is invasive and time-consuming.
- A fast, automated, and non-invasive solution is needed.

Objectives

- Develop a fingerprint-based blood group prediction system.
- Compare deep learning models using accuracy metrics.

Literature Review

1. An Association Between Fingerprint Patterns with Blood Group and Lifestyle Based Diseases: A Review (2021)
- Presented at I2CT 2021, Pune, India.
- Reviewed possible relationship between fingerprint patterns, blood group, and diseases.
- Suggested dermatoglyphics may help in health prediction.

2. A Novel Approach to Predict Blood Group using Fingerprint Map Reading(2025)
- Used fingerprint image processing on dataset of 82 students.
- Applied Multiple Linear Regression (OLS) for prediction.
- Achieved around 62% accuracy.
- Showed low reliability for real-world practical use.

3.Blood Group Detection Using Fingerprint (2025)
- Proposed fingerprint-based blood group prediction system.
- Used image processing and machine learning techniques.
- Showed feasibility of non-invasive blood group detection..

4. Automated Blood Group Detection Using Fingerprint Biometrics and Deep Learning (2025)
- Used 6000+ fingerprint images across 8 blood group classes.
- Hybrid CNN + GNN model achieved 92.4% accuracy.
- Demonstrated strong potential for real-world applications.

Problem Formulation & Deep Learning Technologies

Problem Formulation

- Objective: Predict blood group using fingerprint images
- Task: 8-class image classification
- Input: Fingerprint image
- Output: Predicted blood group
- Challenge: Learning complex fingerprint patterns

Deep Learning Models Used

- LeNet
- VGG16
- ResNet50
- Swin Transformer

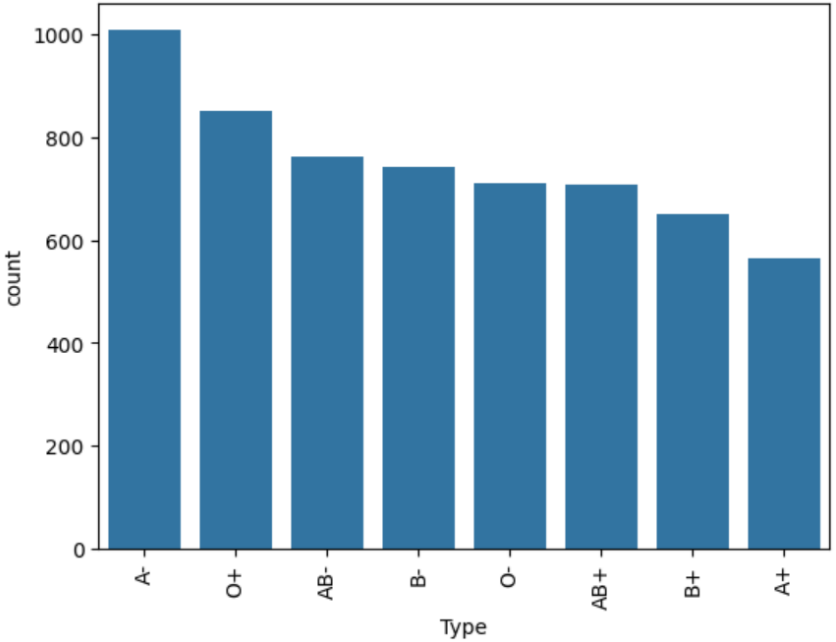
Technology Stack

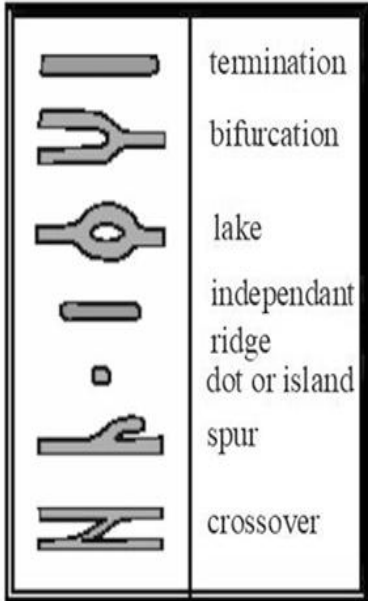
- Core: Python, TensorFlow, Keras
- Processing: NumPy, Pandas, Albumentations
- Visualization: Matplotlib, Seaborn
- Deployment: Streamlit
- Dataset: Kaggle Fingerprint Images

Dataset Description

Dataset from Kaggle

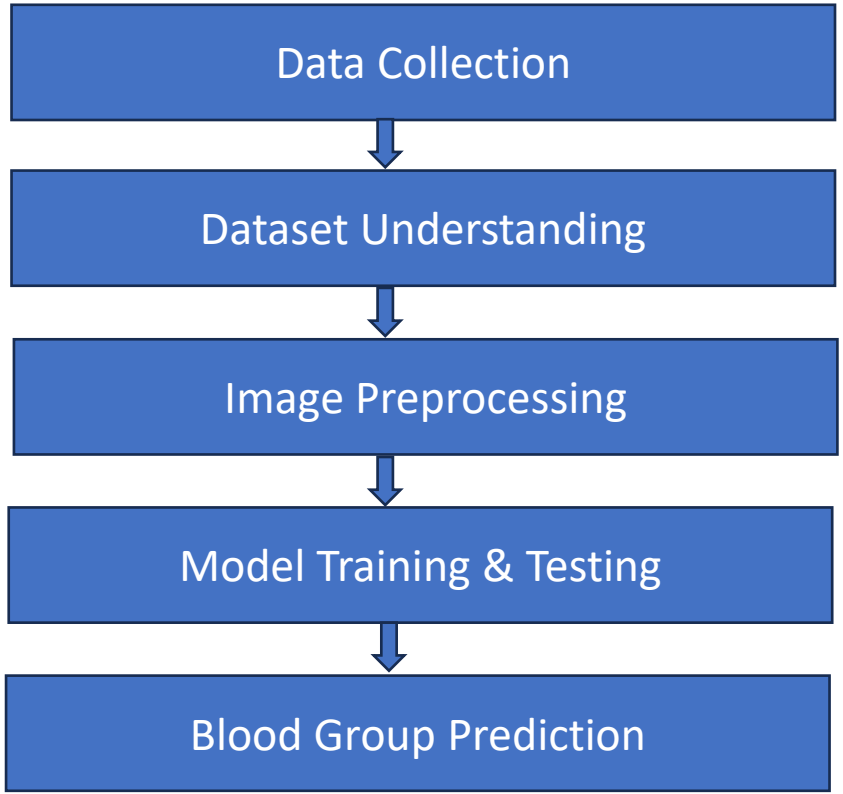
- Approximately 6,000 fingerprint images
- 8 blood group classes







System Workflow



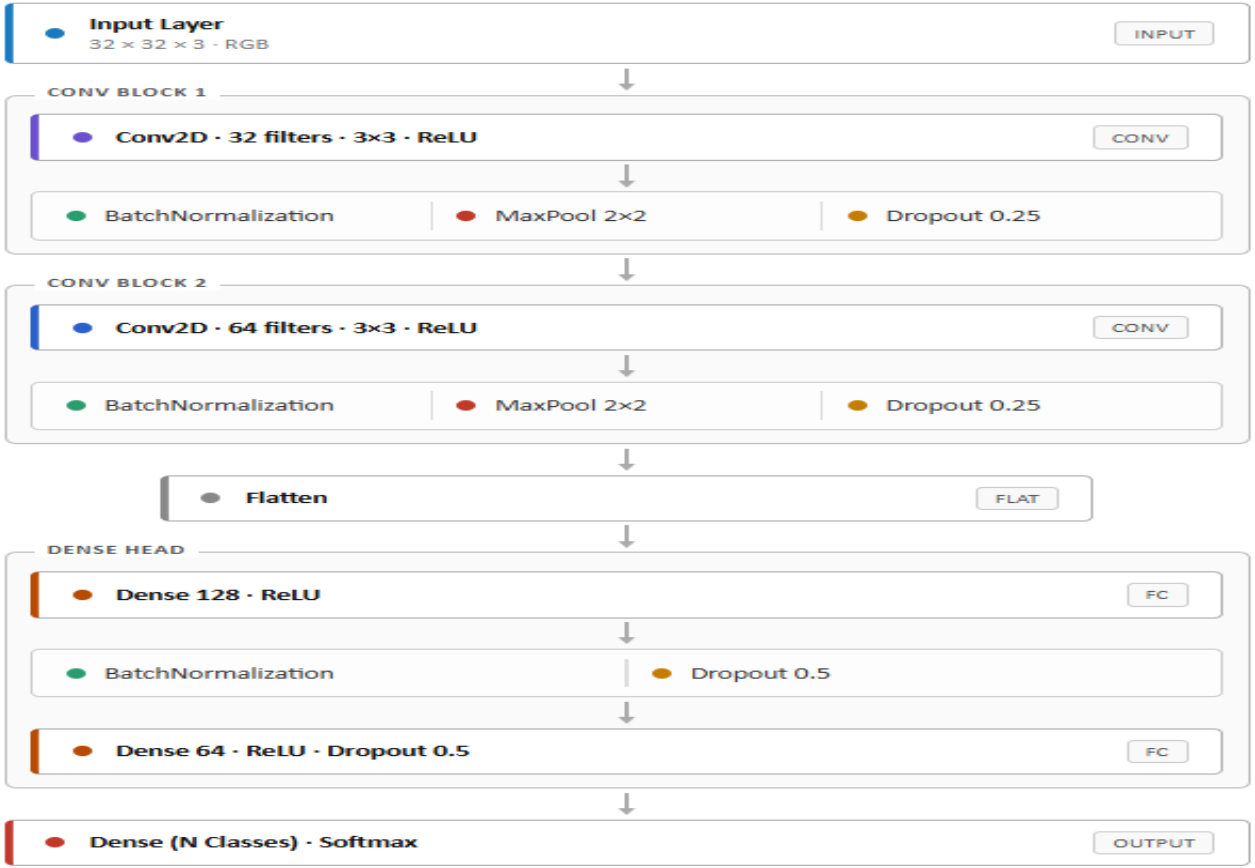
LeNet Architecture & Training Performance

Model Purpose
- Predict blood group from fingerprint images
- 8-class classification

Key Components
- Convolution + ReLU
- Max Pooling
- Fully Connected
- Softmax Output

Training Setup
- TensorFlow / Keras
- Adam Optimizer
- Data Split: 80% Train / 20% Test
- Epochs: 30
- Batch Size: 32 + Augmentation

Performance
- Train Accuracy: 72.71%
- Val Accuracy: 82.92%
- Val Loss: 0.4845



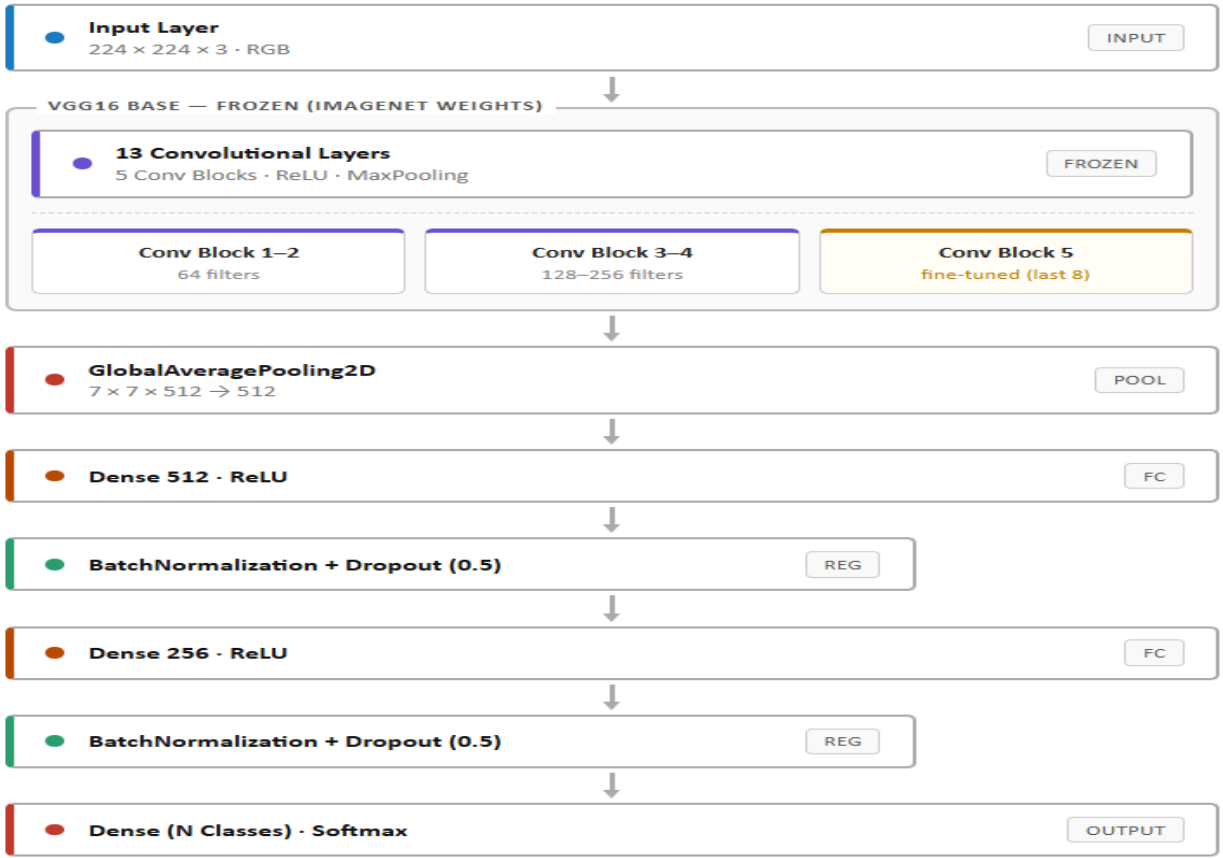
VGG16 Architecture & Training Performance

Model Purpose
- Predict blood group from fingerprint images
- 8-class classification

Key Components
- Pretrained VGG16 Base
- Convolution + ReLU Layers
- Global Average Pooling
- Dense + Softmax Output

Training Setup
- TensorFlow / Keras
- Adam Optimizer (LR: 1e-4)
- Data Split: 80% Train / 20% Test
- Epochs: 30 + 10 Fine-Tuning
- Batch Size: 32 + Transfer Learning

Performance
- Train Accuracy: 73.92%
- Val Accuracy: 87.67%
- Val Loss: 0.3730



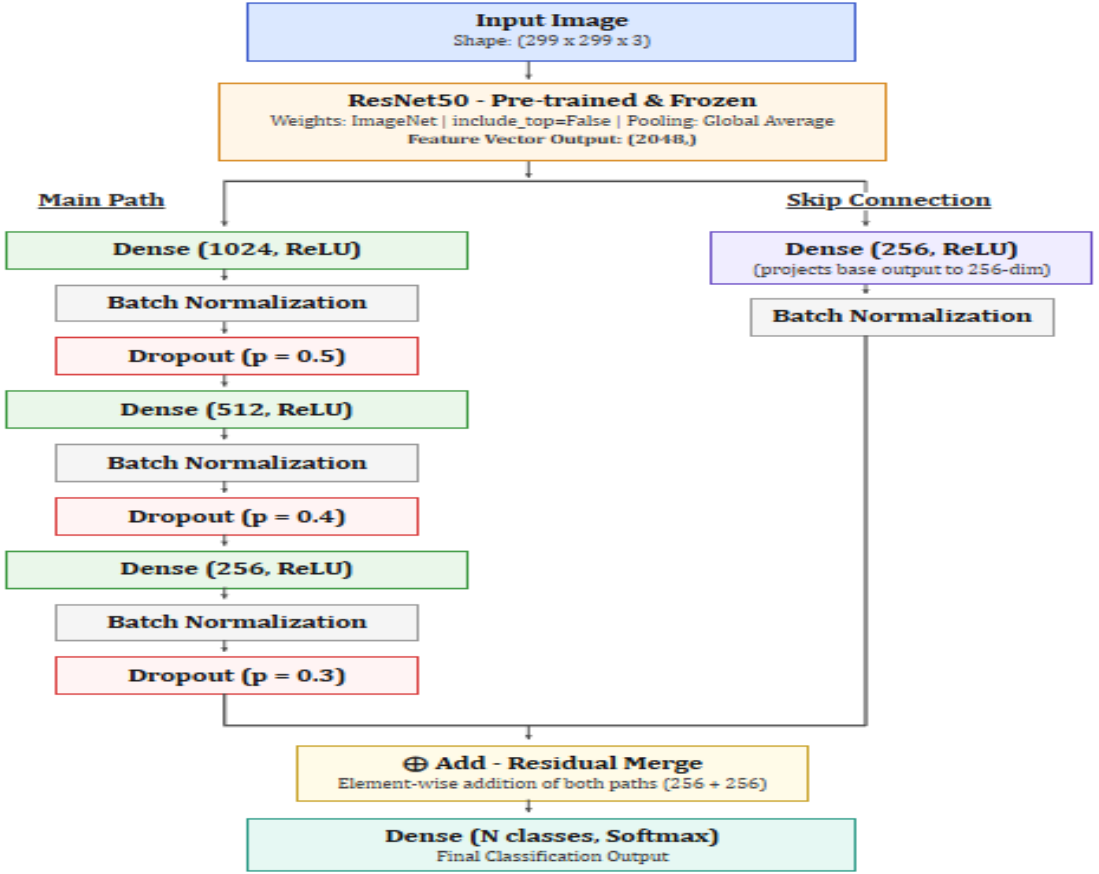
ResNet50 Architecture & Training Performance

Model Purpose
- Predict blood group from fingerprint images
- 8-class classification

Key Components
- Pretrained ResNet50 Base
- Residual Skip Connections
- Global Average Pooling
- Dense + Softmax Output

Training Setup
- TensorFlow / Keras
- Adam Optimizer (Multi-Stage LR)
- Data Split: 80% Train / 20% Test
- Epochs: 65 Total
- Batch Size: 16 + Focal Loss

Performance
- Train Accuracy: 69.04%
- Val Accuracy: 88.67%
- Test Accuracy (TTA): 89.00%
- Validation Loss: 0.0382



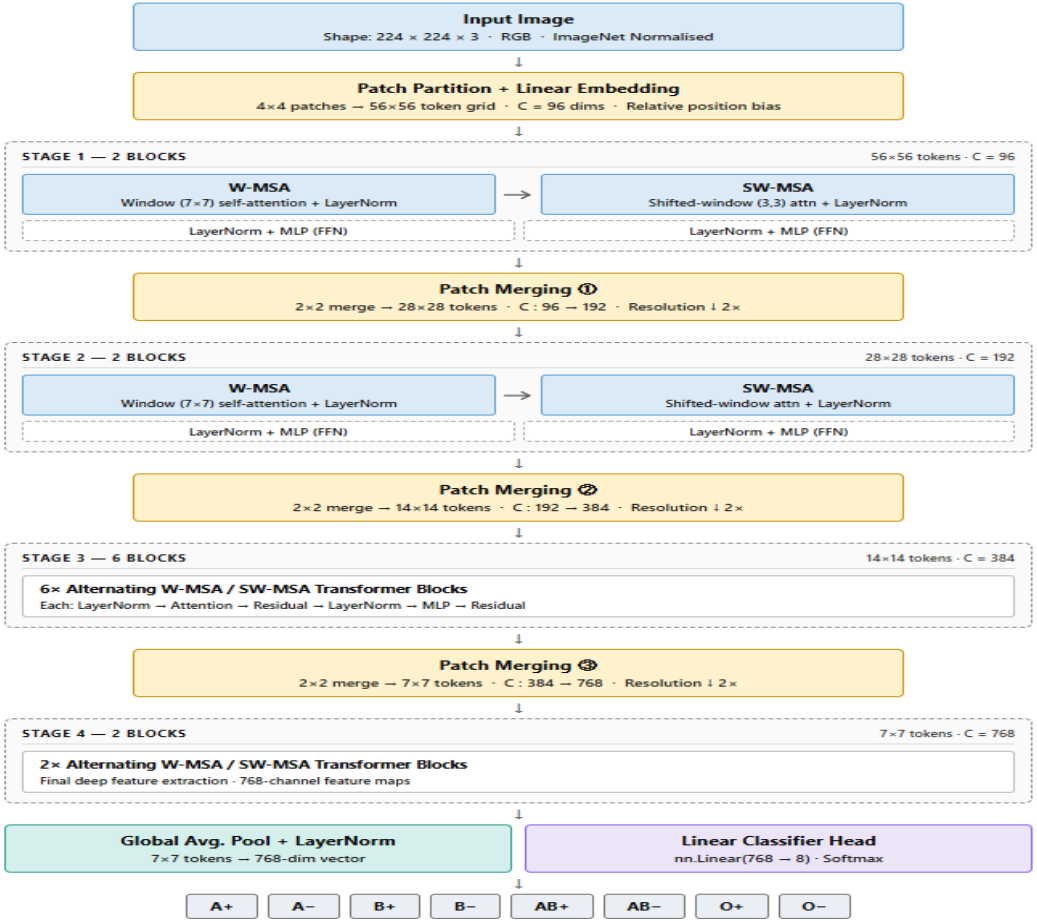
Swin Transformer Architecture & Training Performance

Model Purpose
- Predict blood group from fingerprint images
- 8-class classification

Key Components
- Patch Partition + Embedding
- Window Based Self-Attention
- Shifted Window Attention
- Linear Classifier + Softmax

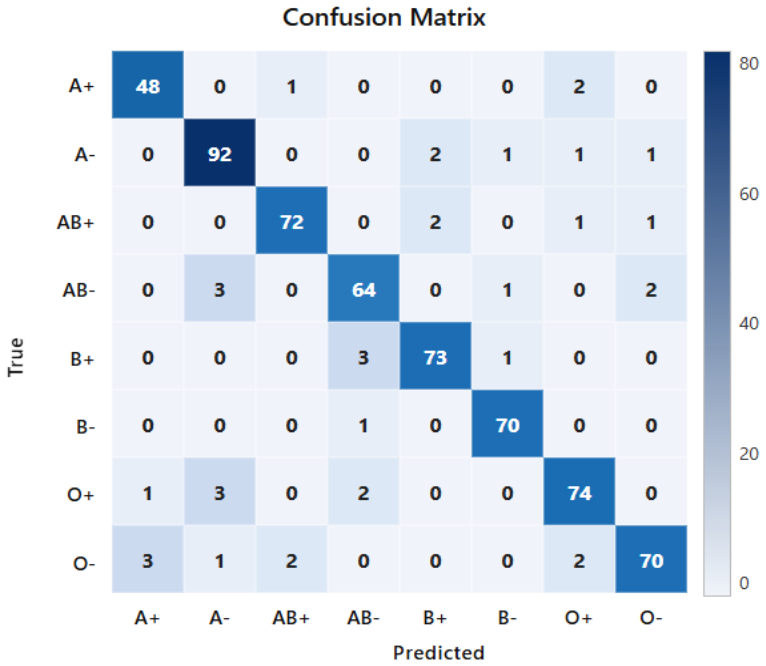
Training Setup
- PyTorch
- AdamW Optimizer (LR: 2e-4)
- Data Split: 80% Train / 10% Val / 10% Test
- Epochs: 12
- Batch Size: 32

Performance
- Train Loss: 0.0267
- Validation Accuracy: 93.59%
- Validation Loss: 0.2114
- Best Performing Model



Performance Comparison of All Models

Model	LeNet	VGG16	ResNet-50	Swim Transformer
Accuracy	82.92%	87.67%	89.00%	93.83%



Classification Report

	precision	recall	f1-score	support
A+	0.92	0.94	0.93	51
A-	0.93	0.95	0.94	97
AB+	0.96	0.95	0.95	76
AB-	0.91	0.91	0.91	70
B+	0.95	0.95	0.95	77
B-	0.96	0.99	0.97	71
O+	0.93	0.93	0.93	80
O-	0.95	0.90	0.92	78
accuracy			**0.94**	**600**
macro avg	**0.94**	**0.94**	**0.94**	**600**
weighted avg	**0.94**	**0.94**	**0.94**	**600**

Conclusion & Future Scope

Conclusion

- Fingerprint-based blood group prediction using deep learning is feasible and effective.
- Models evaluated: LeNet, VGG16, ResNet50, and Swin Transformer.
- Swin Transformer achieved the highest accuracy: 94%.
- The system is non-invasive, fast, and cost-effective.
- Suitable for medical and forensic applications.

Future Scope

- Increase dataset size for better accuracy and generalization.
- Integrate with real-time fingerprint scanners.
- Develop hybrid models (CNN + Transformer).
- Optimize models for mobile and edge devices.
- Perform clinical validation for real-world deployment.

▶ Live Demo

References

1. An Association Between Fingerprint Patterns with Blood Group and Lifestyle Based Diseases: A Review (2021)
https://pubmed.ncbi.nlm.nih.gov/32836652/

2. A Novel Approach to Predict Blood Group Using Fingerprint Map Reading (I2CT 2021)
https://ieeexplore.ieee.org/document/9418090

3. Blood Group Detection Using Fingerprint (Related Research)
https://ijop.net/index.php/mlu/article/view/3669

4. Automated Blood Group Detection Using Fingerprint Biometrics and Deep Learning (2025)
https://arxiv.org/abs/2506.02062

5. Finger Print Based Blood Group Dataset, Kaggle
https://www.kaggle.com/datasets/rajumavinmar/finger-print-based-blood-group-dataset

6. Swin Transformer Research Paper
https://arxiv.org/abs/2103.14030

7. TensorFlow Official Documentation
https://www.tensorflow.org

Thank you